

Session 1 · Deck 1.A1

Who I Am & Why This Course

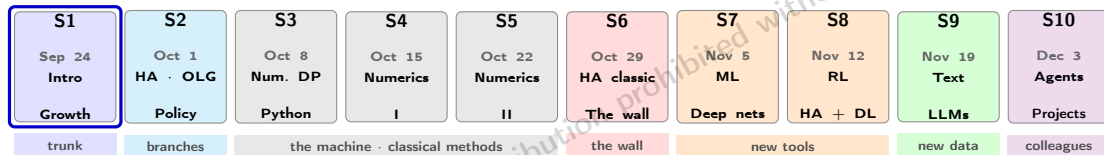
Zhigang Feng

Thursday, September 24, 2026

Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

▼ you are here: 1.A1



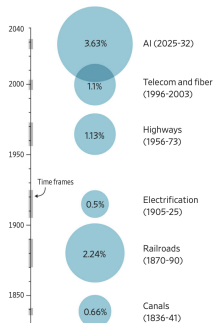
Build it on paper → solve it on the machine → push it to the frontier with AI

Today, Block A: **1.A1** who I am and why this course · **1.A2** the macro frontier · **1.A3** AI as tool and object

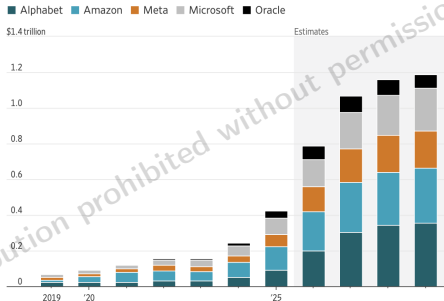
Block B: the optimal growth model — the trunk every later model grows from.

AI Is Here: The Largest Build-Out in U.S. History

Infrastructure spending, % of GDP a year



Capital spending at five large hyperscalers

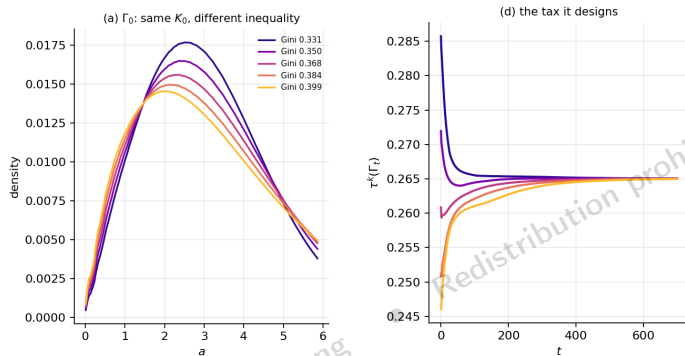


Note: 2026 onward are analyst estimates
Source: FactSet

- **\$10.3 trillion** on data centers and AI infrastructure over 2025–32: **3.6% of GDP a year** — railroads peaked at 2.2%. Never before has the U.S. economy depended so much on the build-out of a single industry.
- Five hyperscalers (Alphabet, Amazon, Meta, Microsoft, Oracle): **\$4.2 trillion** of capex in the four years to 2029, a growing share financed by **debt**.

AI spending is estimated. Spending waves: van Nieuwerburgh (Brookings Institution). Hyperscaler capex: FactSet; 2026 on are analyst estimates.

AI Changes How We Work: A Result We Could Not Compute Before



Five economies with the same aggregate capital $K_0 = 2.454$ and different inequality. The opening capital tax runs from 0.286 at Gini 0.331 down to 0.246 at Gini 0.399 — a spread of 5.7% of the policy cap — and every one converges to the same long-run rate, 0.265.

Research in Practice

The five economies hold **identical aggregate capital**. Only the *distribution* differs — and the government designs a different tax.

The state is a **whole distribution** and the government sets taxes against it — beyond classical methods. Neural-network and RL solvers make it computable.

Feng, Han, Sargent and Zhu (2026) — work in progress.

AI Is Now a Topic of Economics — So We Must Understand How It Works

1. It is here

The build-out runs at **3.6% of GDP a year**, more than any earlier infrastructure wave, and increasingly on debt.

2. How we work

AI does work **beyond our own capacity**: it writes and checks code, reads the literature, turns text into data, and solves models we could not — such as the tax result above.

3. An economic object

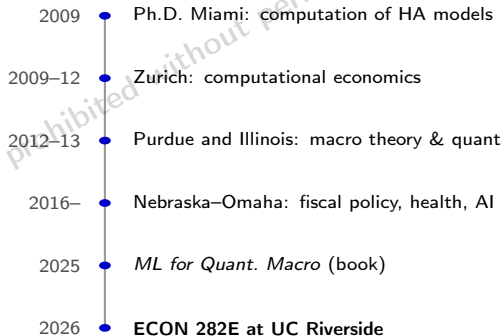
A new kind of capital: it reallocates tasks and wages, must be financed, and invites new policy — questions our theory has to answer.

Hence: we need to understand *how AI works* — to **use its power** in our own research, and to **understand the phenomenon** itself: how it will reshape the economy and economic theory.

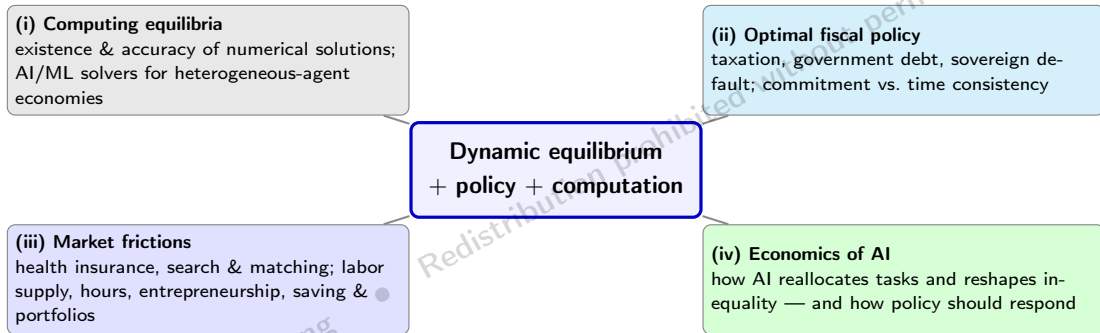
Both sides, tool and object, in depth in 1.A3.

About Me: An Economist Who Computes

- **Training.** Ph.D. Miami (2009) with Manuel Santos and Adrian Peralta-Alva; postdoc at Zurich with Felix Kubler and Karl Schmedders.
- **Positions.** Professor, University of Nebraska–Omaha; earlier Purdue and Illinois; visiting scholar at the St. Louis Fed.
- **Compute.** PI on NSF-ACCESS supercomputing allocations for machine-learning solutions of optimal fiscal policy.
- **Teaching.** Ph.D. macro, recursive methods, quantitative macro, AI/ML for economists; CES–Chow Fellowship (2025).



My Research in One Picture



Every pillar is *also a unit of this course* — you will see the research behind the lectures, including the parts that failed.

Pillar (i): Computing Equilibria

- **A hard question.** With taxes, frictions, or incomplete markets the welfare theorems fail, so we cannot solve a planner's problem. Do our simulations then approximate a true equilibrium?
 - *Numerical Simulation of Non-optimal Dynamic Equilibrium Models* (with Miao, Peralta-Alva & Santos, *IER* 2014).
 - Indeterminacy in overlapping-generations economies (*Math. Methods of OR* 2013; with Hoelle, *Economic Theory* 2017).
- **A new toolkit.** Neural networks and reinforcement learning for economies whose state includes a *whole distribution*:
 - Reinforcement learning for optimal taxation with incomplete markets (with Han & Zhu).
 - Optimal taxation with heterogeneous agents and aggregate uncertainty (with Han, Sargent & Zhu).

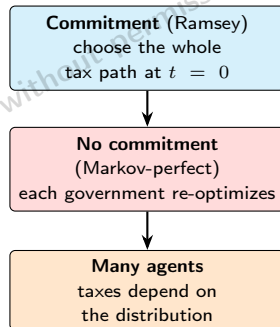
Research in Practice

GRAM — *Generative Reasoning and Augmented Modeling for Quantitative Macroeconomics* (with Miao, Xie, Zhong & Zhou; work in progress). A *typed* knowledge graph over the macro literature, searched under the conditions a result requires: it returns a method **together with the assumptions under which that method is valid**. Scored against no retrieval, chunked retrieval and an untyped graph, with the language model held fixed.

Pillar (ii): Optimal Fiscal Policy — Taxes, Debt, and Commitment

The question: *a government that cannot commit to future taxes — what should it tax today, and how much should it borrow?*

- Time-consistent optimal fiscal policy over the business cycle (*Quantitative Economics* 2015).
- Constitutional constraints on capital and labor taxes (with Domínguez; *Macroeconomic Dynamics* 2017, *Dynamic Games & Applications* 2016).
- Determinants of the national debt; sovereign default and fiscal policy (with Santos).
- Ramsey taxation, public debt, and asset prices with incomplete markets (with Han & Zhu), and its heterogeneous-agent extension (with Chien & Wen).



In this course: Ramsey & time consistency (S2) → why heterogeneity makes it hard (S6) → a neural solution (S8).

Pillar (iii): Frictions — Health Insurance, Work, Entrepreneurship, Saving

The question: *in the U.S. health insurance is tied to your job. What does that do to who works, how much, who starts a business, and who saves?*

- Health, health insurance, and inequality (with Chen & Gu, *International Economic Review* 2025).
- Employment-based health insurance and misallocation (with Chivers & Villamil, *Review of Economic Dynamics* 2017).
- Uncertain medical expenses and aggregate labor supply (with Zhao, *JEBO* 2018).
- Tax subsidies to employer insurance and entrepreneurial talent (with Villamil, *Economic Theory* 2022).
- Outside options and entrepreneurship (with Chivers, Keller & Li), under review at the *International Economic Review*; government debt and intergenerational risk (in progress).

The common engine. A Bewley–Aiyagari economy with extra *states* (health, job, business) and extra *choices* (hours, insurance, occupation, portfolio).

Each friction adds a dimension — which is why computation keeps coming back.

In this course: the engine (S2) and how to solve it (S6).

Pillar (iv): The Economics of AI — Who Gains, and What Should Policy Do?

A task-based model. Each task combines AI services and human labor. When AI gets cheaper, human labor demand in task z moves by

$$\Lambda(z) = \underbrace{\sigma_T s_K(z)}_{\text{task expands}} - \underbrace{\sigma(z) s_K(z)}_{\text{AI replaces labor}}$$

where $s_K(z)$ is AI's cost share, σ_T the substitutability *across* tasks, $\sigma(z)$ *within* the task.

- The **ordering** of Λ across tasks — not its sign — moves workers and wages.
- Rising $\Lambda(z) \Rightarrow$ every high-to-low wage ratio rises, Gini rises.
- A trough in the middle $\Rightarrow$ polarization and U-shaped wage growth.

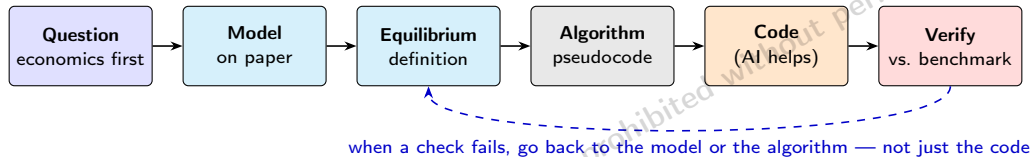
Research in Practice

Working papers and notes (2026).

- *AI, Task Reallocation, and Wage Inequality.*
- *AI for Economics and the Economics of AI (R&R, Digital Economy and Sustainable Development):* what AI does to an occupation turns less on *how much* usage it attracts than on *where inside* it that usage lands.
- Measuring where AI is used, task by task (Economic Index $\times$ O*NET); and taxing compute — an *ad valorem* tax on tokens delays capability rather than walling it off.

In this course: AI as an economic object (1.A3) and measuring it with language models (S9).

Behind the Scenes: What a Research Project Actually Looks Like



Example: taxation without commitment, many agents

1. Classical Aiyagari reference: stationary distribution, bisection on r .
2. Neural solver with no government, graded against (1).
3. Only then: the government enters.

In practice **most of the time goes into verification**: benchmarks, accuracy checks, and asking whether a number makes economic sense.

Writing It Down: *Machine Learning for Quantitative Macroeconomics*



《机器学习与数量宏观经济学：以PyTorch为工具》

Machine Learning for Quantitative Macroeconomics: Using PyTorch

Zhigang Feng — Peking University Press, 2025

- From dynamic programming and classical numerical methods to neural-network and reinforcement-learning solutions of macro models — with code.
- Grew out of a decade of Ph.D. quantitative-macro teaching; free lecture videos and slides accompany it.
- **In this course:** the backbone of S3–S8 (computation, numerical methods, deep learning and RL for macro).

Three Books in Progress

Book	What it covers
《面向经济学家的人工智能》 <i>AI for Economists</i> with Jianjun Miao; Higher Education Press	AI as research infrastructure: machine learning and causal inference, deep learning, reinforcement learning, text and language models, retrieval and agents.
《人工智能与宏观经济学》 <i>AI and Macroeconomics</i> with Jianjun Miao; Higher Education Press	AI as a general-purpose technology: tasks and labor markets, growth, distribution, money and finance, fiscal policy, transition dynamics.
《人工智能经济学导论》 <i>Introduction to AI Economics</i> with Jianjun Miao and Zhe Yuan	The economics of cognition: data, compute and models as inputs; AI in micro, macro, welfare and policy.

Much of this course draws on these drafts — your questions make the books better.

Why This Course, Why Now

1. The questions

The big macro questions — inequality, fiscal sustainability, aging, the effects of AI — are about **distributions** and **policy**.

2. The bottleneck

Once a distribution is a state variable and the government is a strategic player, **computation** is what limits the questions we can ask.

3. The shock

AI changes how we *solve* models, how we *measure* the economy, and what we must *model*: a new kind of capital.

Build it on paper

S1–S2

Solve it on the machine

S3–S6

Push it to the
frontier with AI

S7–S10

By December You Will Be Able To ...

On paper

- ☐ define and characterize equilibrium in the growth, Bewley–Huggett–Aiyagari, Krusell–Smith and OLG models;
- ☐ state a Ramsey problem and explain what goes wrong without commitment.

On the machine

- ☐ solve dynamic programs by VFI and time iteration;
- ☐ choose between perturbation and projection — and defend the choice;
- ☐ solve an Aiyagari economy from scratch.

At the frontier

- ☐ train a neural network on equilibrium conditions and *verify* it;
- ☐ explain when reinforcement learning helps a macroeconomist;
- ☐ use language models as measurement instruments, with validation;
- ☐ run an AI-agent research workflow without losing control of it.

And: present your own project at our mini-conference on **December 3**.

Bridge: From Who to What

- My research asks one kind of question — **dynamic equilibrium with frictions, policy, and heterogeneity** — and keeps running into one constraint: **computation**.
- The course follows the same path: build models on paper, solve them on the machine, and push them to the frontier with AI — *checking every answer along the way*.

Next (1.A2): The macro research frontier.

- Why is wealth far more unequal than income — and what must a model contain to explain it?
- The family tree of workhorse models, the questions each one answers, and why every branch makes computation harder.